

AYJR (April) - Morning 2025 By: C I P H E R

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1: (Only One Option Correct Type)

Five letter words, having distinct letters, are to be constructed using the letters of the word 'EQUATION' so that each word contains exactly three vowels and two consonants. How many of them have all the vowels together?

- (A) 3600
- (B) 1800
- (C) 1080
- (D) 900

Question No. 2: (Only One Option Correct Type)

Consider the system of linear equation
$$\begin{aligned} x - 2y + bz &= 3 \\ ax + 2z &= 2 \\ 5x + 2y &= 1 \end{aligned}$$
 If $ab = 12$ and $a \neq 3$ then system of linear equations has -

- (A) no solution
- (B) infinite solution
- (C) unique solution
- (D) finitely many solutions

Question No. 3: (Only One Option Correct Type)

Let $P(S)$ denote the power set of $S = \{1, 2, 3, \dots, 10\}$. Define the relations R_1 and R_2 on $P(S)$ as AR_1B if $(A \cap B^c) \cup (B \cap A^c) = \emptyset$ and AR_2B if $A \cup B^c = B \cup A^c, \forall A, B \in P(S)$. Then :

- (A) both R_1 and R_2 are equivalence relations
- (B) only R_1 is an equivalence relation
- (C) only R_2 is an equivalence relation
- (D) both R_1 and R_2 are not equivalence relations

Question No. 4: (Only One Option Correct Type)

If the area of the region $\{(x, y) \in R^2 : y^2 \geq 4x, |y| \leq \frac{x}{2} + 2\}$ is Δ , then find the value of $[\Delta]$. [Note: $[S]$ denotes greatest integer less than or equal to S .]

- (A) 12
- (B) 13
- (C) 10
- (D) 11

Question No. 5: (Only One Option Correct Type)

From the focus $(-5, 0)$ of the ellipse $\frac{x^2}{45} + \frac{y^2}{20} = 1$, a ray of light is sent which makes an angle $\cos^{-1}\left(\frac{-1}{\sqrt{5}}\right)$ with the positive direction of x -axis, upon reaching the ellipse surface, the ray is reflected from it. Slope of the reflected ray is

- (A) $-\frac{3}{2}$
- (B) $-\frac{7}{3}$
- (C) $-\frac{5}{4}$
- (D) $-\frac{2}{11}$

Question No. 6: (Only One Option Correct Type)

If the point $(\alpha, 0)$ lies inside the quadrilateral formed by lines $2x + 5y = 15$, $5x - 4y = 21$, $3x + 5y + 17 = 0$ and $y = x + 3$, then which of the following is true?

- (A) Number of prime value(s) of α is 4 .
- (B) Number of integral value(s) of α is 7.
- (C) Minimum integral value of α is -3.
- (D) Maximum integral value of α is 5.

Question No. 7: (Only One Option Correct Type)

In a $\triangle ABC$, the sides BC , CA and AB are consecutive positive integers in increasing order. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ are position vectors of the vertices A , B and C respectively. If $(\vec{c} - \vec{a}) \cdot (\vec{b} - \vec{c}) = 0$, then the value of $|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$ is equal to

- (A) 15
- (B) 16
- (C) 12
- (D) 9

Question No. 8: (Only One Option Correct Type)

If $xdy - y \log_e y dx + 2yx^3 \sin x dx + yx^4 \cos x dx = 0$, then solution is

- (A) $y = e^{x(c-x^2 \tan x)}$
- (B) $y = e^{x^2(c-x \sin x)}$
- (C) $y = e^{x(c+x^2 \sin x)}$
- (D) $y = e^{x^2(c+x \sin x)}$

Question No. 9: (Only One Option Correct Type)

If $a_1, a_2, a_3, \dots, a_{20}$ are the arithmetic means between 13 and 67, then the maximum value of the product $a_1 \cdot a_2 \cdot a_3 \cdot \dots \cdot a_{20}$ is

- (A) $(20)^{20}$
- (B) $(60)^{20}$
- (C) $(80)^{20}$
- (D) $(40)^{20}$

Question No. 10: (Only One Option Correct Type)

The number of points of non-differentiability of the function $f(x) = \max(\sin x, 2x) + [\max(\sin x, 2x)]$ (where $[\cdot]$ denotes greatest integer function) in $(0, 2\pi)$ is

- (A) 12
- (B) 14
- (C) 17
- (D) 18

Question No. 11: (Only One Option Correct Type)

Three fair coins numbered 1 and 0 are tossed simultaneously. Then variance $\text{Var}(X)$ of the probability distribution of random variable X , where X is the sum of numbers on the uppermost faces, is

- (A) 0.7
- (B) 0.75
- (C) 0.65
- (D) 0.62

Question No. 12: (Only One Option Correct Type)

If $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ where $x, y, z \in \{0, 3, -3, 6, -6\}$, then the number of non-zero vectors $\vec{a}$ such that $\vec{a} \cdot \vec{b} = 0$ are

- (A) 20
- (B) 18
- (C) 15
- (D) 12

Question No. 13: (Only One Option Correct Type)

Let $f : R \rightarrow R$ be a polynomial function satisfying the equation $f(f(x) - 2y) = 2x - 3y + f(f(y) - x)$, $\forall x, y \in R$, then the value of $f(9) - f(3)$ is equal to

- (A) 5
- (B) 4
- (C) 6
- (D) 7

Question No. 14: (Only One Option Correct Type)

The sequence a_n is defined by $a_1 = \frac{1}{2}$, $a_{n+1} = a_n^2 + a_n$. Also, $s = \frac{1}{a_1+1} + \frac{1}{a_2+1} + \dots + \frac{1}{a_{100}+1}$ then $[S]$ (where $[\cdot]$ denotes the greatest integer function) is

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Question No. 15: (Only One Option Correct Type)

A square matrix P satisfies $P^2 = I - 2P$ where I is the identity matrix if $P^2 + P^3 + P^4 = a^2I - b^2P$ then mark incorrect option

- (A) $a^2 + b^4$ is a perfect square
- (B) $a^4 + b^2$ is a perfect square
- (C) $a^2 + b^2$ is prime
- (D) $a^4 + b^4$ is prime

Question No. 16: (Only One Option Correct Type)

Let k be the coefficient of x^{18} in the polynomial $f(x) = (1+x)^{20} + x(1+x)^{19} + x^2(1+x)^{18} + \dots + x^{18}(1+x)^2$ then the value of $\frac{k}{190}$ is equal to _____.

- (A) 8
- (B) 7
- (C) 6
- (D) 5

Question No. 17: (Only One Option Correct Type)

Let $f : (0, \infty) \rightarrow R$ be a differentiable function satisfying $f(x) + e^{f(x)} = \frac{2}{x} - \ln x - 1$. Find the number of integers in the range of x satisfying the inequality $f(2x^2 + 1) - f(x^2 + 5) \geq f(1), x > 0$.

- (A) 5
- (B) 3
- (C) 2
- (D) 1

Question No. 18: (Only One Option Correct Type)

If a, b and c are real numbers such that $a^2 + b^2 + c^2 - ab - bc - ac \leq 0$, then

$$\begin{vmatrix} (a-b+1)^5 & b^7 - c^7 & c^9 - a^9 \\ a^{11} - b^{11} & (b-c+2)^3 & c^{13} - a^{13} \\ a^{15} - b^{15} & b^{17} - c^{17} & (c-a+3)^1 \end{vmatrix} =$$

- (A) $2abc$
- (B) 0
- (C) $24abc$
- (D) 24

Question No. 19: (Only One Option Correct Type)

The locus of the point (x, y) whose distance from the line $y = 2x + 2$ is equal to the distance from $(2, 0)$, is a parabola with the length of latus rectum same as that of the parabola $y = Kx^2$, then the value of K is equal to

- (A) $\frac{\sqrt{5}}{12}$
- (B) $\frac{\sqrt{5}}{4}$

- (C) $\frac{4}{\sqrt{5}}$
 (D) $\frac{12}{\sqrt{5}}$

Question No. 20: (Only One Option Correct Type)

The value of the expression $\tan\left(\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{2}{9}\right) + \tan^{-1}\left(\frac{1}{8}\right) + \tan^{-1}\left(\frac{2}{25}\right) + \tan^{-1}\left(\frac{1}{18}\right) + \dots \infty\right)$, is:

- (A) 2
 (B) 3
 (C) 4
 (D) 5

Mathematics Numerical (Maximum Marks: 20)

Question No. 21: (Numerical Type)

The shortest distance between the following pair of lines: $\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$ and $\vec{r} = 3\hat{i} + 3\hat{j} - 5\hat{k} + \mu(2\hat{i} + 3\hat{j} + 6\hat{k})$ is $\frac{\sqrt{293}}{K}$. Find the value of K .

Question No. 22: (Numerical Type)

For some function $f(x)$ and $g(x)$ which are differentiable $\forall x > 0$ satisfy the following condition. (i) $\left(\frac{f(x)}{x}\right)' = x^2 e^{-x^2}$ (ii) $g(x) = \frac{4}{e^4} \int_1^x e^{t^2} \cdot f(t) dt$ (iii) $f(1) = \frac{1}{e}$ Find the value of $3e^4(f(2) - g(2))$.

Question No. 23: (Numerical Type)

If $x dy = y(dx + y dy)$, $y(1) = 1$, $y(x) > 0$, then $y(-3)$ is

Question No. 24: (Numerical Type)

Let $f: A \rightarrow B$ be any function where A is a set containing the positive integral solution of the inequality $\operatorname{cosec}^{-1}(\operatorname{cosec} 2) > x^2 - 3x$ and B is the set of all divisors of the natural number 2010. If $f(i) \leq f(j) \forall i < j$, then find the total number of mappings from A to B .

Question No. 25: (Numerical Type)

Let e be the eccentricity of a hyperbola and $f(e)$ be the eccentricity of its conjugate hyperbola. If n is an even positive integer, then $\int_1^3 \underbrace{f \circ f \circ \dots \circ f}_n(e)$ is equal to 'a' then $2a$ is equal to

Physics Single Correct (Maximum Marks: 80)

Question No. 26: (Only One Option Correct Type)

A certain liquid has a melting point of $-50^\circ C$ and a boiling point of $150^\circ C$. A thermometer is designed with this liquid and its melting and boiling points are designated at $0^\circ L$ and $100^\circ L$. The melting and boiling points of water on this scale are

- (A) $25^\circ L$ and $75^\circ L$, respectively
- (B) $0^\circ L$ and $100^\circ L$, respectively
- (C) $20^\circ L$ and $70^\circ L$, respectively
- (D) $30^\circ L$ and $80^\circ L$, respectively

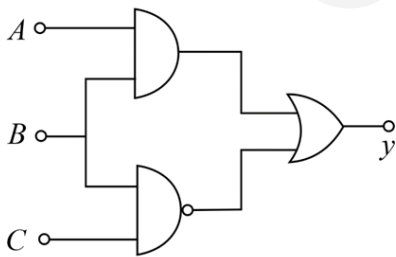
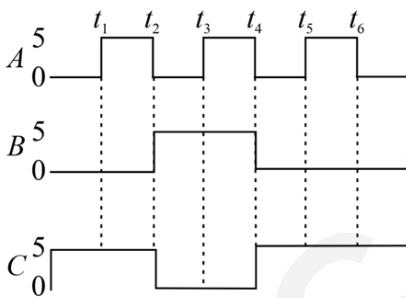
Question No. 27: (Only One Option Correct Type)

There are three persons A, B & C moving with constant velocity. Speed of A is 10 m/sec towards east, velocity of B relative to A is 6 m/sec at an angle of $\cos^{-1}\left(\frac{15}{24}\right)$ north of east. The velocity of C relative to B is 12 m/sec towards west. What will be the magnitude of velocity of C in m/sec ?

- (A) 3
- (B) 5
- (C) 2
- (D) 4

Question No. 28: (Only One Option Correct Type)

For the given circuit, the input digital signals are applied at the terminals A , B and C . What would be the output at the terminal y ?



- (A) 0 V
- (B) 5 V
0 V
- (C) 5 V

(D)  0 V
 5 V

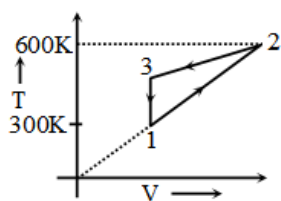
Question No. 29: (Only One Option Correct Type)

In Young's double slit experiment, the separation between two slits is $0.3 \times 10^{-3}\text{ m}$. When light beam is passed through the two slits, then interference pattern is observed on the screen 1.5 m away. If the first violet and red fringes are formed 2.0 mm and 3.5 mm away from the central white fringe, then the difference in wavelength of red and violet light is $x \times 10^{-9}\text{ m}$. The value of x is

- (A) 100
- (B) 300
- (C) 400
- (D) 500

Question No. 30: (Only One Option Correct Type)

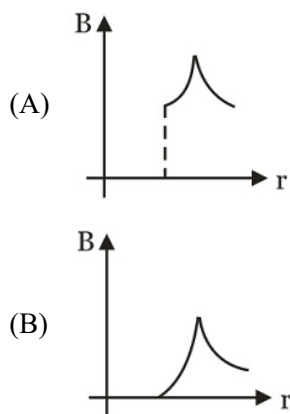
Two moles of an ideal gas have undergone a cyclic process $1 - 2 - 3 - 1$. If net heat exchange in the process is -300 J , then work done by the gas in the process $2 - 3$ is ($R = 8.3\text{ J mol}^{-1}\text{ K}^{-1}$)

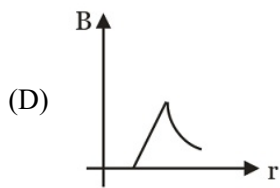
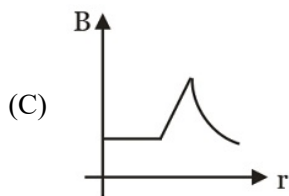


- (A) 5280J
- (B) -4680J
- (C) 4680J
- (D) -5280J

Question No. 31: (Only One Option Correct Type)

A current i is uniformly distributed over the cross-section of a long hollow cylindrical wire of inner radius R_1 and outer radius R_2 . Magnetic field B varies with distance from the axis of the cylinder as





Question No. 32: (Only One Option Correct Type)

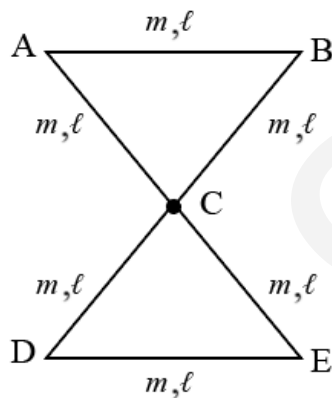
Assertion- The red light is just able to emit photoelectrons from a metal surface, then violet light cannot emit the photoelectrons from that metal.

Reason- The energy of violet light photons is more than red light photons.

- (A) Both Assertion and Reason are true but Reason is not correct explanation of Assertion.
- (B) Assertion is true but Reason is false.
- (C) Assertion is false but Reason is true.
- (D) Both Assertion and Reason are false.

Question No. 33: (Only One Option Correct Type)

An arrangement of rods each of mass m and length l are welded (wherever required) to form a shape as shown. The moment of inertia about an axis passing through point C and perpendicular to the plane of figure is:

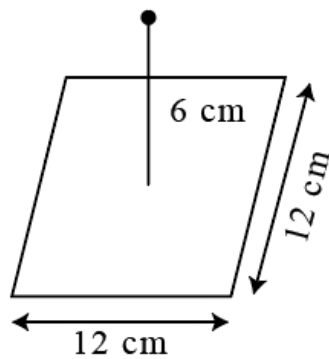


- (A) ml^2
- (B) $\frac{3}{2}ml^2$
- (C) $3ml^2$
- (D) $\frac{9}{2}ml^2$

Question No. 34: (Only One Option Correct Type)

As shown in figure, a point charge $+20 \mu\text{C}$ is placed 6 cm vertically above the center of a square of side 12 cm. As a result of this

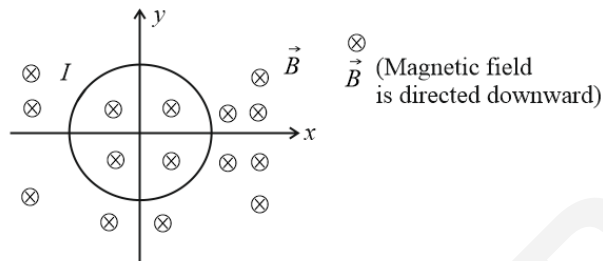
arrangement, the electric flux through the square will be



- (A) $2.5 \times 10^6 \text{ N m}^2 \text{ C}^{-1}$
- (B) $3.8 \times 10^5 \text{ N m}^2 \text{ C}^{-1}$
- (C) $4.2 \times 10^5 \text{ N m}^2 \text{ C}^{-1}$
- (D) $2.9 \times 10^6 \text{ N m}^2 \text{ C}^{-1}$

Question No. 35: (Only One Option Correct Type)

Assertion: The loop shown in the diagram will have tendency to expand.



Reason: Net force acting on a closed loop in external uniform magnetic field is zero.

- (A) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.
- (B) If both Assertion and Reason are true but Reason is not correct explanation of the Assertion.
- (C) If Assertion is true but the Reason is false.
- (D) If Assertion is false but Reason is true.

Question No. 36: (Only One Option Correct Type)

The wavelength of a spectral line emitted by hydrogen atom in the Balmer series is $\frac{16}{3R}$ (R is Rydberg constant). What is the value of the principal quantum number of the state from which the transition takes place?

- (A) 5
- (B) 6
- (C) 3
- (D) 4

Question No. 37: (Only One Option Correct Type)

For the angle of minimum deviation of a prism to be equal to its refracting angle, the prism must be made of a material whose refractive index:

- (A) lies between $\sqrt{2}$ and 1

(B) lies between 2 and $\sqrt{2}$

(C) is less than 1

(D) is greater than 2

Question No. 38: (Only One Option Correct Type)

A small conducting sphere of the radius r is lying concentrically inside a bigger hollow conducting sphere of radius R . The bigger and smaller sphere are charged with Q and q ($Q > q$) and are insulated from each other. The potential difference between the sphere will be

(A) $\frac{1}{4\pi\epsilon_0} \left(\frac{q}{r} - \frac{q}{R} \right)$

(B) $\frac{1}{4\pi\epsilon_0} \left(\frac{q}{R} - \frac{Q}{r} \right)$

(C) $\frac{1}{4\pi\epsilon_0} \left(\frac{q}{r} - \frac{Q}{R} \right)$

(D) $\frac{1}{4\pi\epsilon_0} \left(\frac{Q}{R} + \frac{q}{r} \right)$

Question No. 39: (Only One Option Correct Type)

Frequencies of various radiations are given as

$f_v \rightarrow$ Visible light

$f_r \rightarrow$ Radio waves

$f_{uv} \rightarrow$ Ultra Violet waves

Then which of the following is true?

(A) $f_{uv} < f_v < f_r$

(B) $f_r < f_v < f_{uv}$

(C) $f_v < f_r < f_{uv}$

(D) $f_{uv} < f_r < f_v$

Question No. 40: (Only One Option Correct Type)

A gas satisfies the relation $PV^{5/3} = K$ where P is pressure, V is volume and K is constant. The dimensions of constant K are

(A) $ML^4 T^{-2}$

(B) $ML^2 T^{-2}$

(C) $ML^6 T^{-2}$

(D) MLT^{-2}

Question No. 41: (Only One Option Correct Type)

A cylindrical metallic rod in thermal contact with two reservoirs of heat at its two ends conducts an amount of heat Q in time t . The metallic rod is melted and the material is formed into a rod of half the radius of the original rod. What is the amount of heat conducted by the new rod, when placed in thermal contact with the two reservoirs in time t ?

(A) $\frac{Q}{4}$

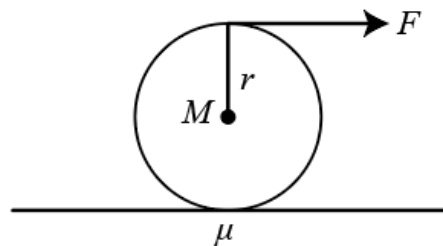
(B) $\frac{Q}{16}$

(C) $2Q$

(D) $\frac{Q}{2}$

Question No. 42: (Only One Option Correct Type)

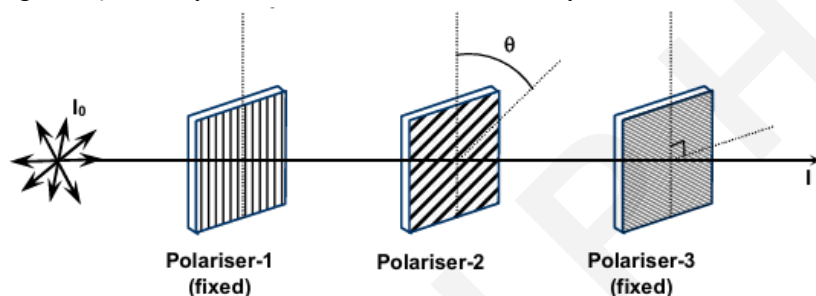
A ring was initially at rest on a rough floor and a force F (parallel to floor) is applied on the top of the rolling body. Choose the correct option for the pure accelerated rolling.



- (A) The kinetic friction (f_k) will act in opposite direction of force
- (B) The static friction must act
- (C) The friction will not act in the backward direction
- (D) If rolling body is a disc, the static friction will act in the backward direction

Question No. 43: (Only One Option Correct Type)

A student writes down four conclusions that he observes while performing a polarization experiment with 3 polarisers as shown in the figure. The leftmost and the rightmost polariser are fixed and crossed and the middle one can be rotated. The intensity of the unpolarised light is I_0 . Identify the incorrect conclusion noted by him.



- (A) If $\theta = 0^\circ$ then $I = 0$
- (B) If $\theta = 90^\circ$ then $I = 0$
- (C) If $\theta = 45^\circ$ then $I = \frac{I_0}{8}$
- (D) If $\theta = 30^\circ$ then $I = \frac{3I_0}{16}$

Question No. 44: (Only One Option Correct Type)

Assertion : It is not possible for a system, unaided by an external agency to transfer heat from a body at lower temperature to another body at higher temperature. Reason : According to Clausius statement, " No process is possible whose sole result is the transfer of heat from a cooled object to a hotter object.

- (A) If both assertion and reason are true and the reason is the correct explanation of the assertion.
- (B) If both assertion and reason are true but reason is not the correct explanation of the assertion.
- (C) If assertion is true but reason is false.
- (D) If the assertion and reason both are false.

Question No. 45: (Only One Option Correct Type)

A ball of mass 100 gm and carrying charge 1 C is released from a point near the surface of earth, in a uniform horizontal magnetic field

of 2T. Find the maximum speed attained by the ball in m/s. (Take $g = 10 \text{ m/s}^2$, assume ball attains maximum speed before hitting ground)

- (A) 1.5 m/s
- (B) 1 m/s
- (C) 2 m/s
- (D) 0.5 m/s

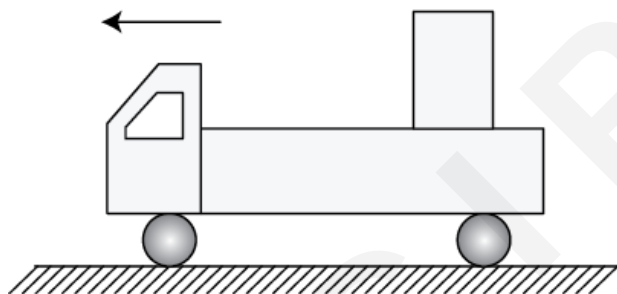
Physics Numerical (Maximum Marks: 20)

Question No. 46: (Numerical Type)

An equiconvex lens made of glass of refractive index $\frac{3}{2}$ has focal length f in air. It is completely immersed in water of refractive index $\frac{4}{3}$. The percentage change in the focal length is

Question No. 47: (Numerical Type)

The rear side of a truck is open and a box of 40 kg mass is placed 5 m away from the open end as shown. The coefficient of friction between the box & the surface below it is 0.15. On a straight road, the truck starts from rest and accelerates with 2 ms^{-2} . At what distance (in m.) from the starting point does the box fall off the truck (i.e. distance travelled by the truck)? [Ignore the size of the box] [value of $g = 10 \text{ m/s}^2$]



Question No. 48: (Numerical Type)

A ball falling in a lake of 200 m shows a decrease of 0.1 % in its volume. The Bulk modulus of elasticity of the material of the ball is $\text{_____} \times 10^9 \text{ N/m}^2$. (take $g = 10 \text{ m/s}^2$)

Question No. 49: (Numerical Type)

An ideal choke takes a current of 10amp when connected to an AC supply of 125 volt and 50 Hz. A pure resistor under the same conditions takes a current of 12.5amp. If the two are connected to an AC supply $100\sqrt{2}$ volt and 40 Hz, then find the current in series combination of above resistor and inductor. If it is $5 \times n$ Amp. Find value of n

Question No. 50: (Numerical Type)

The escape velocity from a planet is v_0 . The escape velocity from a planet having twice the radius but same density is nv_0 then n is:

Chemistry Single Correct (Maximum Marks: 80)

Question No. 51: (Only One Option Correct Type)

Which of the following statements are correct?

1. Electron density in xy plane in $3d_{x^2-y^2}$ orbital is zero.
2. Electron density in xy plane in $3d_{z^2}$ orbital is zero.
3. 2s orbital has only one spherical node in it.
4. For $2p_x$ orbital yz is the nodal plane.

- (A) 3 and 4
(B) All of these
(C) 2
(D) 1 and 3

Question No. 52: (Only One Option Correct Type)

Which of the following does not represent property stated against it?

- (A) $\text{Co}^{2+} < \text{Fe}^{2+} < \text{Mn}^{2+}$ - Ionic size
(B) $\text{Ti} < \text{V} < \text{Mn}$ - Number of oxidation states
(C) $\text{Cr}^{2+} < \text{Mn}^{2+} < \text{Fe}^{2+}$ - Paramagnetic behaviour
(D) $\text{Sc} < \text{Cr} < \text{Fe}$ - Density

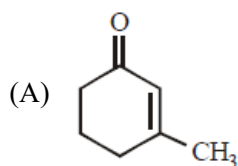
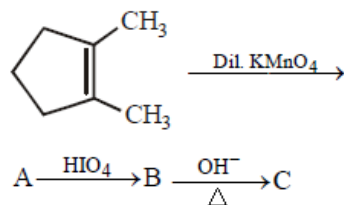
Question No. 53: (Only One Option Correct Type)

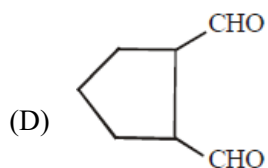
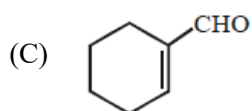
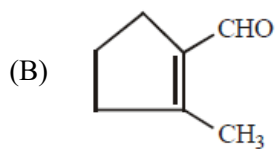
Which of the following pairs have different hybridisation and same shape? (i) NO_3^- and CO_3^{2-} (ii) SO_2 and NH_2^- (iii) XeF_2 and CO_2 (iv) H_2O and NH_3

- (A) (i) and (iv)
(B) (ii) and (iv)
(C) (ii) and (iii)
(D) None of these

Question No. 54: (Only One Option Correct Type)

What will be the correct structural formula of the final product for the following reaction?





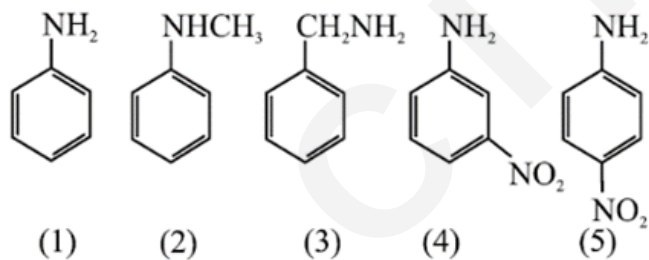
Question No. 55: (Only One Option Correct Type)

Amongst the following statements I : The structure of $[\text{BrF}_4]^+$ is regular tetrahedral II: Bond order of O_2 decreases by the removal of an electron III : Br_3^- and Br_3^+ are having same shape and same structure IV : The hybrid orbital of phosphorous in PCl_5 is $\text{sp}^3 \text{d}_{xy}$ The incorrect statements is/are

- (A) only II
(B) I, III, IV
(C) I, II, III
(D) I, II, III, IV

Question No. 56: (Only One Option Correct Type)

Which order for basic character of amine is correct for following compounds?



- (A) $3 > 1 > 2 > 5 > 4$
(B) $3 > 2 > 1 > 5 > 4$
(C) $3 > 1 > 2 > 4 > 5$
(D) $3 > 2 > 1 > 4 > 5$

Question No. 57: (Only One Option Correct Type)

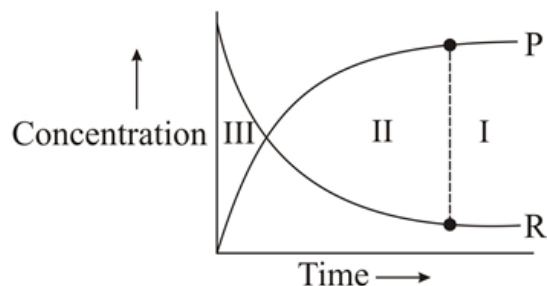
The $E_{\text{M}^{3+}/\text{M}^{2+}}$ values for Cr, Mn, Fe and Co are -0.41 , $+1.57$, $+0.77$ and $+1.97$ V respectively. For which one of these metals the change in oxidation state from +2 to +3 is easiest?

- (A) Cr
(B) Mn
(C) Fe

(D) Co

Question No. 58: (Only One Option Correct Type)

For the reaction $R \rightleftharpoons P$, variation of concentration is plotted against time. The time at which the equilibrium establishes is as shown:



Which of the following regions show(s) equilibrium?

- (A) III
- (B) II
- (C) I
- (D) Both II and III

Question No. 59: (Only One Option Correct Type)

Prussian blue is a deep blue pigment containing Fe^{2+} , Fe^{3+} and CN^- ions. It has the formula $\text{Fe}_7(\text{CN})_{18}$. How many Fe^{2+} and Fe^{3+} ions are there per formula unit?

- (A) 3Fe^{2+} and 4Fe^{3+}
- (B) 4Fe^{2+} and 3Fe^{3+}
- (C) 5Fe^{2+} and 2Fe^{3+}
- (D) 6Fe^{2+} and 1Fe^{3+}

Question No. 60: (Only One Option Correct Type)

An organic compound contains 49.3% carbon, 6.84% hydrogen and its vapour density is 73. Molecular formula of the compound is :

- (A) $\text{C}_3\text{H}_5\text{O}_2$
- (B) $\text{C}_4\text{H}_{10}\text{O}_2$
- (C) $\text{C}_6\text{H}_{10}\text{O}_4$
- (D) $\text{C}_3\text{H}_{10}\text{O}_2$

Question No. 61: (Only One Option Correct Type)

Wavelength of red light is absorbed by the complex

- (A) $[\text{Cu}(\text{CN})_4]^{2-}$
- (B) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- (C) CuSO_4

(D) $\text{Cu}(\text{CN})_2$.

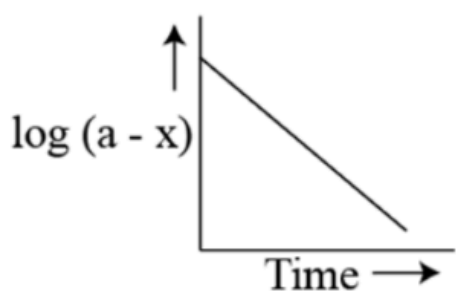
Question No. 62: (Only One Option Correct Type)

Assertion : Molecular nitrogen is less reactive than molecular oxygen. Reason : The bond length of N_2 is shorter than that of oxygen.

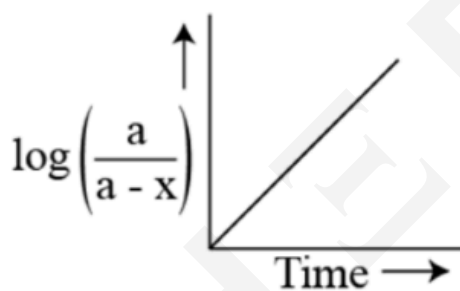
- (A) If both assertion and reason are true and reason is the correct explanation of assertion.
- (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (C) If assertion is true but reason is false.
- (D) If both assertion and reason are false.

Question No. 63: (Only One Option Correct Type)

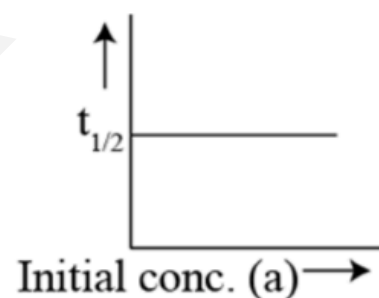
Which of the following is/are correct for the first order reaction? (a is initial concentration of reactant, x is concentration of the reactant reacted and t is time)



(I)



(II)



(III)

- (A) I and II only
- (B) II and III only
- (C) I, II and III only
- (D) I and III only

Question No. 64: (Only One Option Correct Type)

Which of the following reaction is used to produce aldehyde?

- (A) $\text{CH}_3\text{CH}_2\text{COOH} \xrightarrow[\text{ii) H}_3\text{O}^+]{\text{i) LiAlH}_4/\text{ether}}$
- (B) $\text{CH}_3\text{COCl} \xrightarrow{\text{R}_2\text{CuLi}}$
- (C) $\text{CH}_3\text{COOH (vapours)} \xrightarrow[\Delta]{\text{MnO}}$
- (D) $\text{CH}_3\text{CN} \xrightarrow[\text{ii) H}_3\text{O}^+]{\text{i) SnCl}_2/\text{HCl}}$

Question No. 65: (Only One Option Correct Type)

K_a for HCN is 5×10^{-10} at 25°C . For maintaining a constant pH of 9, the volume of 5 M KCN solution required to be added to 10 mL of 2 M HCN solution is

- (A) 4 mL
- (B) 8 mL

(C) 2 mL

(D) 9 mL

Question No. 66: (Only One Option Correct Type)

Assertion : ionization potential across the period is $\text{Na} < \text{Al} < \text{Mg} < \text{Si}$. Reason : ionization potential decreases with decrease in atomic size.

(A) If both assertion and reason are true and the reason is the correct explanation of the assertion.

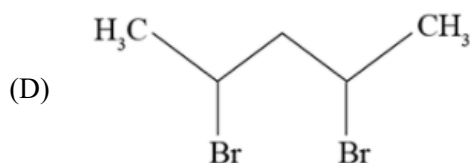
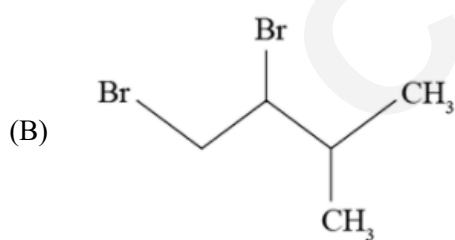
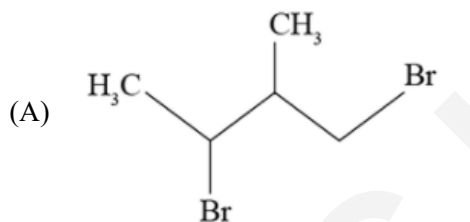
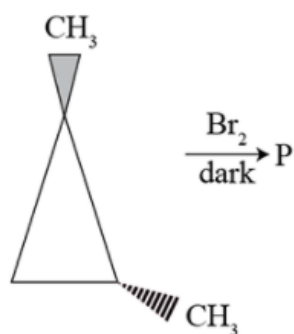
(B) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(C) If assertion is true but reason is false.

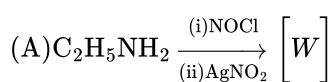
(D) If the assertion and reason both are false.

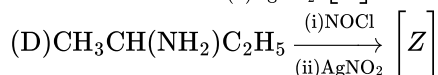
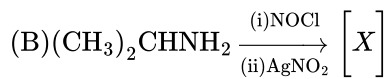
Question No. 67: (Only One Option Correct Type)

The major product P of the following reaction is



Question No. 68: (Only One Option Correct Type)





Which product will not show tautomerism?

- (A) *W*
 (B) *X*
 (C) *Y*
 (D) *Z*

Question No. 69: (Only One Option Correct Type)

Compare vitamin **List I** with its deficiency disease **List II**.

	Column-I		Column-II
(A)	Vitamin - B ₁₂	(1)	Sterility
(B)	Vitamin - B ₆	(2)	Haemorrhagic condition
(C)	Vitamin - E	(3)	Pernicious anaemic
(D)	Vitamin - K	(4)	Skin diseases

Codes :

- (A) A B C D
 1 2 3 4
 (B) A B C D
 2 3 4 1
 (C) A B C D
 3 4 1 2
 (D) A B C D
 3 4 2 1

Question No. 70: (Only One Option Correct Type)

The enthalpies of formation of N₂O and NO are 28 and 90 kJ mol⁻¹ respectively. The enthalpy of the reaction 2N₂O(g) + O₂(g) → 4NO(g) is equal to

- (A) 8 kJ
 (B) 88 kJ
 (C) -16 kJ
 (D) 304 kJ

Chemistry Numerical (Maximum Marks: 20)

Question No. 71: (Numerical Type)

In Carius method for estimation of halogens, 0.2 g of an organic compound gave 0.188 g of AgBr. The percentage of bromine in the compound is _____. (Nearest integer)

[Atomic mass: Ag = 108, Br = 80]

Question No. 72: (Numerical Type)

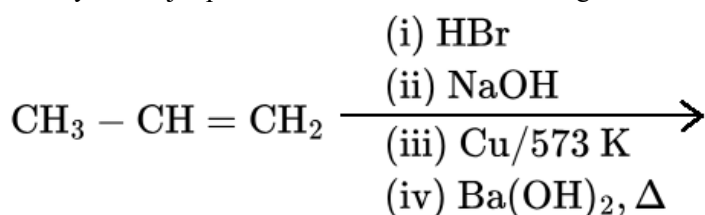
Ethylene glycol is used as an antifreeze in cold climate. Mass of ethylene glycol which should be added to 4 kg of water to prevent it from freezing at -6°C will be g. (K_f for water = $1.86\text{ K kg mol}^{-1}$ and molar mass of ethylene glycol = 62 g mol^{-1})

Question No. 73: (Numerical Type)

For the two parallel reaction $A \xrightarrow{k_1=2\text{ sec}^{-1}} B$ and $A \xrightarrow{k_2=4\text{ sec}^{-1}} C$, the activation energy E' for the disappearance of A is given in terms of activation energies E_1 and E_2 for the two path by $E' = \frac{E_1+2E_2}{x}$. The value of x is _____.

Question No. 74: (Numerical Type)

Identify the major product formed from the following



How many carbon - hydrogen bond pairs are there in final product?

Question No. 75: (Numerical Type)

V_1 mL of NaOH of molarity X and V_2 mL of Ba(OH)_2 of molarity Y are mixed together. The mixture is completely neutralized by 100 mL of 0.1 N HCl. If $V_1/V_2 = 1/4$ and $X/Y = 4$ then the fraction of acid is neutralized by Ba(OH)_2 is K, find 3K. (nearest integer).
